

Chapter 7: Clustering

7.5 Clustering in Non-Euclidean Spaces

7.6 Clustering for Streams and Parallelism

Course: **Advanced Big Data Analysis**

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Introduction

The Challenge

Traditional clustering algorithms often assume:

- Data exists in Euclidean space
- Distances can be measured using straight lines
- All data points are available in memory

Real-world scenarios often involve:

- Complex, non-Euclidean relationships
- Streaming data that evolves over time
- Massive datasets exceeding memory limits

Solutions

GRGPF Algorithm:

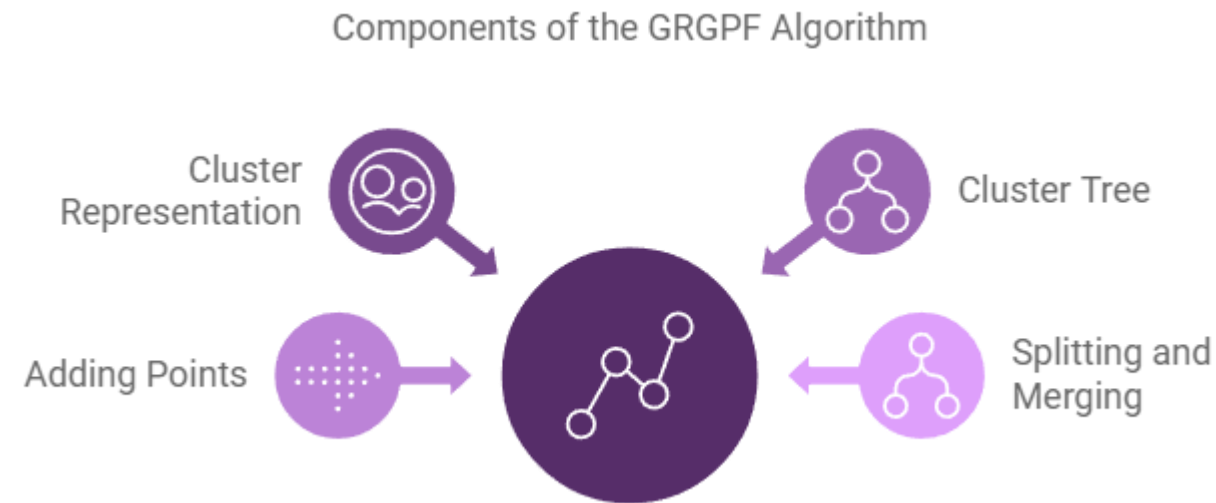
- Works in any distance-based space
- Maintains cluster representations in main memory
- Uses tree structure for efficient organization

Key Benefits:

- Handles non-main-memory data efficiently
- Adapts to various data spaces and relationships
- Supports real-time clustering updates

The GRGPF Algorithm

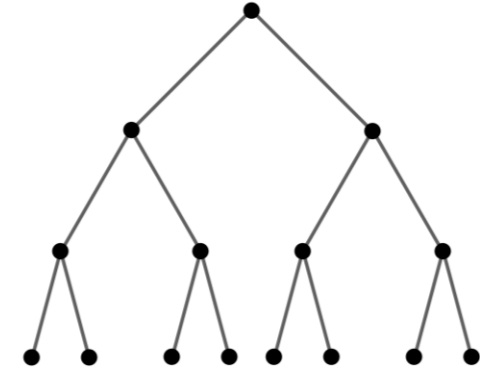
- Designed for Non-Euclidean spaces
- Ideas: Mixed approach:
 - Point-assignment
 - Organizes clusters hierarchically



The GRGPF Algorithm - Cluster Representation

- Let $ROWSUM(p)$ be the sum of the *squares of the distances* from p to each of the other points in its cluster
- Clusters are represented with features:
 - N , the number of points in the cluster
 - The ***clustroid*** of the cluster (the point that minimizes $ROWSUM$)
 - The k points closest to the ***clustroid***
 - The k points furthest from the ***clustroid***

The GRGPF Algorithm - Tree Structure



- Organizes clusters hierarchically, similar to B-trees or R-trees
- “closer” leaves contains closer clusters
- **Leaf Nodes:** Store detailed representations of a limited number of clusters.
- **Interior Nodes:** Hold a fixed-size sample of clustroids from descendant clusters and pointers to subtrees.

The GRGPF Algorithm - Adding Points

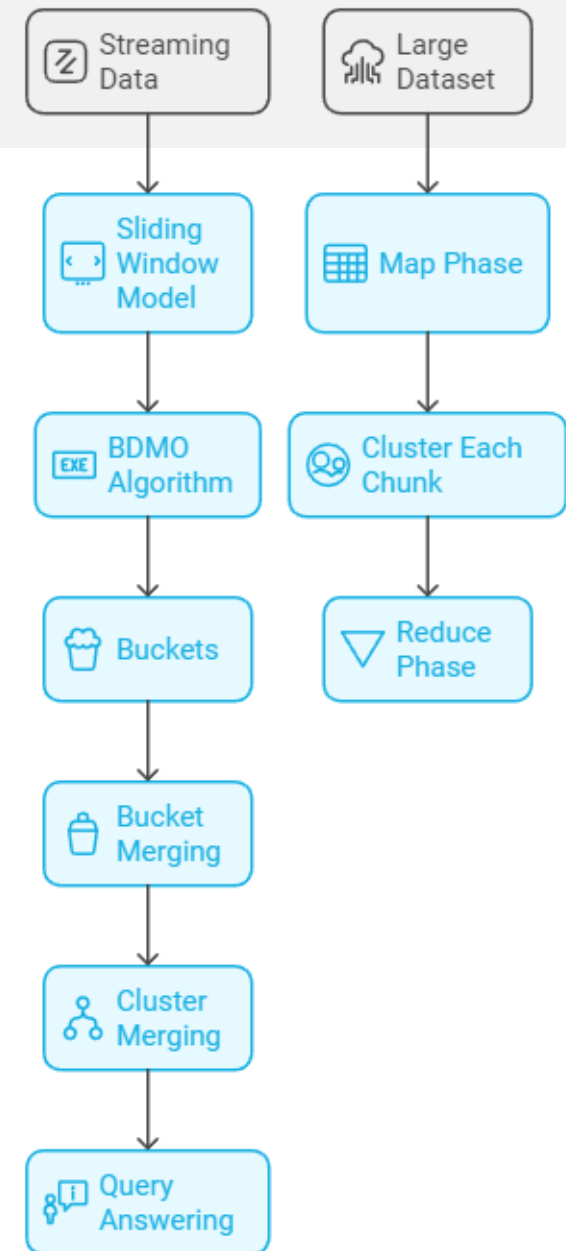
- Point Assignment:
 - New points are inserted into the cluster with the nearest clustroid.
 - The tree is traversed from the root down, using sample clustroids at each level to guide the search.
 - The point is assigned to the cluster at the leaf node whose **clustroid** exhibits the smallest distance to the point.
- Dynamic Update: The addition of a new point triggers updates to the chosen cluster's representation:
 - Increment the cluster's point count (N).
 - Update the *ROWSUM* of all representative points (including the clustroid, nearest points, and furthest points) in the cluster representation to incorporate the new point p .
$$ROWSUM(p) = ROWSUM(c) + Nd^2(p, c)$$
 - Estimate the *ROWSUM* of the new point, potentially incorporating it into the cluster's representation if it becomes one of the k -nearest or k -furthest points from the clustroid

The GRGPF Algorithm - Splitting and Merging Cluster

- **Radius Limit** = $\sqrt{ROWSUM(c)/N}$
- **Splitting overgrown Clusters:** Clusters are split when their radius exceeds a limit.
- **Merging Nearby Clusters:** If the cluster-representing tree grows too large for main memory, GRGPF initiates the merging of nearby clusters to reduce the tree's size. This situation typically arises when the initial radius limit is set too low.
$$ROWSUM_C(p) = ROWSUM_{C_1}(p) + N_{C_2}(d^2(p, c_1) + d^2(c_1, c_2)) + ROWSUM_{C_2}(c_2)$$
- **Merging Considerations:** Clusters are considered for merging based on proximity, often those at the same leaf node or with a common parent node. Merging involves selecting a new clustroid from distant points of the merging clusters and updating the cluster representation's features.

Clustering for Streams and Parallelism

- **Challenges in Stream Clustering:** Involves clustering data that arrives continuously and evolves over time, requiring algorithms that can adapt to changing patterns.
- **Parallelism for Large Datasets:** For large datasets, parallel clustering techniques can be used to improve efficiency. This often involves distributing the data across multiple machines and performing clustering in parallel.



Clustering for Streams

- The Sliding Window Model: Commonly used for stream clustering. It focuses on the most recent N data points, providing a snapshot of the current data distribution.
- Query on last $m \leq N$ points
- Clusters change over time: Stream clustering algorithms must handle changes in cluster patterns, including migrating centroids, emerging clusters, and merging clusters

Clustering for Streams - The BDMO Algorithm

- Bucketing Strategy: It partitions the stream into buckets of sizes that are powers of two. The bucket sizes increase as we go back in time. Thus, every p stream elements, we create a new bucket, with the most recent p points.
- Bucket Contents: Each bucket stores *summaries* of clusters formed from the points falling within the bucket's time range. These *summaries* include information such as:
 - The number of points in the cluster
 - The centroid or clustroid of the cluster
 - Additional parameters for merging decisions (e.g., diameter or radius)

Clustering for Streams - Merging Buckets in BDMO

- K-Means in Euclidean Space

- Cluster Representation: Clusters are represented by their centroids and the number of points they contain.
- Matching: Clusters are paired based on the distance between their centroids.
- Formula: The centroid c of a merged cluster is calculated as a weighted average of the centroids of the merged clusters:
 - $c = (n_1c_1 + n_2c_2) / (n_1 + n_2)$
- where:
 - c_1 and c_2 are the centroids of the clusters being merged.
 - n_1 and n_2 are the number of points in the clusters being merged.

- Non-Euclidean Space (GRGPF Techniques)

- Cluster Representation: Clusters are represented by clustroids and rowsums. The rowsum is the sum of the squares of the distances from each point in the cluster to the clustroid.

Clustering in a Parallel Environment

- Map Stage:
 - Data is divided into chunks and distributed to multiple Map tasks.
 - Each Map task performs clustering on its assigned chunk of data.
 - The output of each Map task is a set of key-value pairs, where the key is typically a fixed value and the value is a cluster representation.
- Reduce Stage:
 - A single Reduce task receives the cluster representations from all Map tasks.
 - The Reduce task merges the local clusters into a final global set of clusters.
 - The merging strategy can vary depending on the clustering objectives

Thank you